

Skills Summary

Solving Systems of Two Linear Equations by Graphing

Use this reference while completing your worksheet or when studying.

A system asks for the pair (x, y) that makes *both* equations true at once. Graphed, each equation is a line and each point on it is one pair that works, so the pair that works for both is exactly where the two lines cross.

1. Graphing from slope-intercept form

Rule: in $y = mx + b$, plot the intercept $(0, b)$ first, then use $m = \frac{\text{rise}}{\text{run}}$ to step to a second point and draw the line. Do this for both equations; the crossing point is the solution.

Always check by substituting the point into both equations — a graph read to the nearest square can be a square off.

Example: $y = x - 1$ and $y = -2x + 5$

Step 1. Both are already solved for y , so plot each intercept and step off each slope; no rearranging is needed.

Step 2. From $(0, -1)$ step up 1, right 1. From $(0, 5)$ step down 2, right 1. The lines cross once, and substituting confirms it: $1 = 2 - 1$ and $1 = -4 + 5$.

(2, 1)

Try it: $y = 2x - 4$ and $y = -x + 5$

(2, 3) check

2. Graphing from standard form

Rule: for $Ax + By = C$, the fastest two points are the intercepts. Set $x = 0$ and solve for y ; set $y = 0$ and solve for x . Plot both and join them.

If the intercepts land off the grid, rearrange into $y = mx + b$ instead — the two methods give the same line.

Example: $x + 2y = 6$ and $x - y = 3$

Step 1. Neither equation is solved for y , and both have small whole-number intercepts, so the intercept method is quicker than rearranging.

Step 2. $x + 2y = 6$ passes through $(0, 3)$ and $(6, 0)$; $x - y = 3$ passes through $(0, -3)$ and $(3, 0)$. Substituting the crossing point gives $4 + 2 = 6$ and $4 - 1 = 3$.

(4, 1)

Try it: $2x + y = 6$ and $x - y = 3$

(0, 3) check

3. When the lines do not cross once

Same slope, different intercept: the lines are parallel, never meet, and the system has **no solution**.

Same slope and same intercept: the two equations describe one line drawn twice, so every point on it works — **infinitely many solutions**. Check for this by simplifying one equation; if it turns into the other, they are the same line.

Example: $y = 2x + 3$ and $y = 2x - 1$

Step 1. Compare the slopes before graphing: both are 2, so the lines rise identically and can only be parallel or identical.

Step 2. The intercepts differ (3 and -1), so the lines stay a fixed distance apart. Setting the right sides equal gives $3 = -1$, which is impossible.
no solution

Try it: $y = -x + 5$ and $2x + 2y = 10$. How many solutions?

check: one line, $y + x = 5$, so infinitely many

4. Reading the crossing point in a story

Rule: write each situation as $y = (\text{rate})x + (\text{start})$, graph both, and read the crossing point. The x -coordinate says *when* the two are equal and the y -coordinate says *what* they are both equal to.

Before the crossing point the line that started higher is on top; after it, the steeper line is.

Example: Gym A charges 12 dollars to join plus 3 dollars a month; Gym B charges 6 to join plus 5 a month. When do they cost the same?

Step 1. Each cost is a start value plus a repeated rate, so both are lines: $y = 3x + 12$ and $y = 5x + 6$.

Step 2. Gym A starts higher but climbs more slowly, so B catches up. At the crossing point $3(3) + 12 = 21$ and $5(3) + 6 = 21$.

(3, 21) — equal at month 3, at 21 dollars

Try it: $y = 2x + 10$ and $y = 4x + 4$. When are they equal, and at what value?

check: (3, 16)